Reading Robustness from One Truncated Trace: Per-Model White-Box Recipes for Spurious Math Reasoning

Sabyasachi Choudhury
IIT Kanpur
sabyasachi25@iitk.ac.in

Garv
IIT Kanpur
garv25@iitk.ac.in

Abstract

The AIMO Interpretability Challenge asks whether a reasoning model's success on a competition math problem survives faithful rewordings of the problem. We predict this from a single greedy reasoning trace truncated at 6144 tokens, with white-box access to the model. Our method calls every problem whose reasoning finishes within the budget robust, and scores unfinished traces with separate methods per model: a rank ensemble of self-verification share, answer survival when the reasoning is cut, hidden-state sensitivity to programmatic rewordings, attention look-back and forced-answer confidence. for Qwen3.5-4B, we also use the entropy of the forced answer after 4096 tokens, and for Olmo-3-7B-Think, a Min-K% familiarity router and a second model acting as a judge. Our two submissions differ only in Olmo's unfinished calls: the second replaces them with an early-trace hidden-state direction. They score 1.00 and 0.93 on the public leaderboard (all-robust: 0.50) and make 33 and 36 correct calls on the official training labels for our four models (43 rows; all-robust: 31). Removing the output-distribution signals lowers leaderboard accuracy to 0.71–0.79; removing the hidden-state and attention signals leaves the first submission unchanged, while the second's hidden-state direction loses three training set points, while losing one on the leaderboard. We also report a negative result: labels we generated with our own perturbations do not reproduce the official labels, which shaped how we validated everything. Code: `https://github.com/RKResident/AIMO-Interp-Challenge`.

1 Introduction

Both nominated submissions share one test-time pipeline (Section 3) and differ only in how unfinished Olmo problems are called. Both methods are covered here. S1 (`permodel-v8-ent4096`) is the base method. S2 (`permodel-v8-ent4096-olmoseg25`) is S1 with one single-feature override for Olmo (Section 3.5).

Contributions. A pipeline that given a model and a problem, generates one greedy trace and a few teacher forced passes, and predicts whether the model was solving the problem robustly or not. We also include ablation studies and our reports on the various white-box methods we tried on this task

2 Data and evaluation

Models. Qwen3.5-4B [Qwen Team, 2026], Skywork-OR1-Math-7B [He et al., 2025a], Olmo-3-7B-Think [Team Olmo, 2025] and DeepSeek-R1-0528-Qwen3-8B [DeepSeek-AI, 2025, Yang et al., 2025]; perturbation robustness in the sense of Mirzadeh et al. [2025], labelled by the challenge [Stefánik et al., 2026].

Preprint.

Official labels. MAIN: the organisers' training set `train-main-v2`, restricted to our four models: 43 rows (Qwen 12, Skywork 12, Olmo 9, DeepSeek 10), 31 robust. This is the only labelled data we evaluate on; we report correct calls out of 43 with Wilson 95% intervals [Wilson, 1927]. **Public leaderboard.** We report accuracy on the leaderboard's 14 problems. With 3-5 problems per model its estimates are also very noisy.

Our own data, used for calibration only. Unlabelled greedy traces for $\sim$100 post-cutoff competition problems per model (POOL) supply the reference distributions and median thresholds below, which need no labels. We also labelled problems ourselves (released with the code). No official threshold for the accuracy drop was published, so we inferred the rule from the data. In MAIN, every robust row has a worst-case drop ≤ 0.10 and every non-robust row ≥ 0.25, with nothing in between, and the same rule, counting never-solved problems as robust, reproduces the labels of the public validation sample exactly. We used 8 temperature-1 samples per version and five perturbation types, four written by Claude Sonnet and one (typos) scripted. Every generation of our own was capped at 6144 new tokens, matching the test-time trace, with answers forced at the cap when the model had not finished (the relabel in Section 6 used 32k); the official labels come from a much larger generation budget. These labels are used only for low-capacity choices: the sign of each feature, the per-model feature subsets and, for S2, one direction. Section 6 shows that they are noisy and track difficulty, so every such choice was kept only if it also held on the official labels, and no accuracy below is measured on our labels (the "ours" column of Table 4 is a diagnostic).

Regime split. At a 6144-token cap most MAIN traces are still thinking (32/43). All 11 MAIN rows that finish are robust. Every non-robust label therefore falls on an unfinished trace, which is where every recipe acts.

3 Method

Trace. We build the challenge prompt ("Solve the following mathematical problem step by step. Put your final answer within `\boxed{}`") with each model's chat template. We decode greedily (bf16, HF Transformers [Wolf et al., 2020]) up to 6144 new tokens, stopping early on end-of-thinking, end-of-sequence or a 3$\times$32-token loop. **Finished** (the end-of-thinking token appears) $\Rightarrow$ robust. **Unfinished** $\Rightarrow$ score with the model's recipe.

All features are computed on that one trace (definitions in Appendix A). We group them by what they read; Section 5.1 ablates the groups.

3.1 Text features (black-box)

Finished rule: whether the end-of-thinking token appears within the cap. `chk`: the share of $\geq$4-word sentences of the reasoning that check, verify or confirm something.

3.2 Output-distribution features (grey-box)

These read the model's next-token distribution under teacher forcing. `fa`: the top-1 probability of the first answer token after appending "`</think>` The final answer is `$\boxed{`" at 6144 tokens. H_b: the entropy of that distribution when the trace is cut at b tokens. *Answer survival* [cf. early answering, Lanham et al., 2023]: `nrk` and `f1k`, the probability of the clean forced answer when the reasoning is removed entirely or cut to its first 1024 tokens, relative to the clean probability. `cfl`: the change in the trace's log-likelihood when it is teacher-forced after three automatic rewordings of the problem (10% typos, a neutral distractor sentence, renamed variables). Min-K% [Shi et al., 2024] ($k = 20$): the mean of the lowest 20% token log-probabilities of the problem text, a familiarity score.

3.3 Internal features (white-box)

Hidden states: `cfc`, the cosine between late-trace hidden states of the original pass and of the three reworded passes; the Olmo direction of S2 (Section 3.5). *Attention*: `lble` and `pmle`, the Lookback-Lens ratio [Chuang et al., 2024] and the attention mass on the prompt, each as last quarter minus first quarter of the trace; `jmp`, the attention-weighted mean backward distance between 64-token blocks (robust traces keep re-reading their own setup). The Qwen judge used for Olmo (below) is also internal, since Qwen's ensemble consists of `lble`, `pmle`, `jmp` and `cfc`.

Table 1: Correct calls on MAIN (43 rows) with Wilson 95% intervals, and public leaderboard accuracy (14 problems). On MAIN, Olmo's judge needs a Qwen trace of each Olmo problem, which exists for 2 of the 4 unfinished Olmo rows; there it makes the same calls as the router, so Olmo's S1 calls use the router branch. S1 is best on the leaderboard, S2 on MAIN; on MAIN both gains over all-robust are inside the intervals.

Method	MAIN correct (43)	Leaderboard score
All robust (baseline)	31 [0.57, 0.83]	0.50
Finished ⇒ R, unfinished ⇒ NR	23 [0.39, 0.67]	0.57
S1 (`v8-ent4096`)	33 [0.62, 0.87]	**1.00**
S2 (S1 + Olmo direction)	**36** [0.70, 0.92]	0.93

3.4 From features to calls

Rank-ensemble score. For model m with feature subset F_m, a value x_f is mapped to its percentile q_f among the reference values $R_{m,f}$ (POOL traces plus our labelled traces, unfinished only):

$$s = \frac{1}{|F_m|} \sum_{f \in F_m} \left(\tilde{q}_f - \tfrac{1}{2}\right), \qquad \tilde{q}_f = \begin{cases} q_f & \sigma_f = +1 \\ 1 - q_f & \sigma_f = -1 \end{cases}, \qquad \text{robust} \iff s > \tau_m. \tag{1}$$

τ_m is the median score s of model m's calibration traces. The signs σ_f are the direction of each feature's length-controlled AUC. The subsets F_m were chosen per model by subset search on our labelled rows: Qwen {`lble, pmle, jmp, cfc`} (Qwen's own unfinished calls use H_{4096} instead, below; this ensemble serves as the judge for Olmo); Skywork {`fa, chk, f1k`}; DeepSeek {`chk, nrk, cfl, pmle`}. Percentile scoring makes the score insensitive to feature scale and to shifts between our hardware and the evaluator's.

Olmo: novelty router and judge. Olmo's features flipped direction between problems it had seen in training and post-cutoff problems. We route on Min-K%: below -5.93 (the cut that best separated post-cutoff from older problems in POOL, using no robustness labels) goes to a novel branch {`f1k`}, otherwise to a familiar branch {`chk, lble, pmle`}, each with its own median threshold. A judge then loads Qwen, generates Qwen's own trace of the same problem and scores it with Qwen's recipe. Olmo's call is robust iff the mean of two label-free percentiles exceeds 0.5: Olmo's margin among its POOL margins, and Qwen's margin among Qwen's margins on the same POOL problems. If the judge fails or time is short, Olmo's own call stands.

Qwen: forced-answer entropy at 4096 tokens. For unfinished Qwen problems, S1 replaces the rank-ensemble call with H_{4096} (applied last, after any judging): robust iff $H_{4096} < 1.621$, the median over Qwen's 100 POOL traces. The sign comes from our 66 labelled Qwen problems (AUC 0.77 for lower-is-robust). Qwen's rank-ensemble margins are still computed, because the Olmo judge uses them.

3.5 S2: an early-trace hidden-state direction for Olmo

S2 runs the full S1 pipeline and then, after the judge, replaces the call only for unfinished Olmo problems. The router and judge are only a fallback for Olmo in S2. We take the mean hidden state over the 256 tokens around 25% of the trace, at the 5th decoder layer, and project it onto a precomputed unit robust-minus-spurious difference-of-means direction fitted on our 21 labelled unfinished Olmo problems. Robust iff the projection is above 0.234, the median over Olmo's 50 POOL traces. For both single-feature readouts the threshold is label-free and the sign comes from our data. Choosing which feature to use for these two models, however, used MAIN results and leaderboard scores of earlier builds (see Limitations).

4 Results

Table 1 gives the headline numbers. On the leaderboard S1 scores 1.00 and S2 0.93, against 0.50 for all-robust. Both leaderboard scores are upper bounds rather than estimates of private-set accuracy: the recipes, including S1's Qwen rule and S2's Olmo direction, were chosen with the leaderboard scores of earlier builds in view. MAIN is the less biased measure, and there the gains are small: it

Table 2: Component ablation: correct calls per model on MAIN (Qwen 12, Skywork 12, Olmo 9, DeepSeek 10 rows; of them 11/7/4/10 unfinished) and leaderboard accuracy (LB). The per-model recipes help DeepSeek and Skywork most; the Qwen entropy helps on both sets; the Olmo direction helps on MAIN but costs one leaderboard problem.

Variant	Qwen	Skywork	Olmo	DeepSeek	Total	LB
All robust	8	9	7	7	31	0.50
Finished only (unfinished NR)	5	8	7	3	23	0.57
v8 (rank ensembles, no overrides)	8	10	5	9	32	0.93
v8 without Olmo router/judge	8	10	7	9	34	0.79
S1 = v8 + Qwen H_{4096}	9	10	5	9	33	1.00
v8 + Olmo direction only	8	10	8	9	35	0.86
S2 = S1 + Olmo direction	9	10	8	9	36	0.93

Table 3: White-box ablation by feature group. *No internals*: every hidden-state and attention feature, the Olmo direction and the Qwen judge removed; the two submissions then coincide, since they differ only in an internal readout. *No output*: `fa`, `nrk`, `f1k`, `cfl`, the Min-K% router and H_{4096} removed (Qwen falls back to its internal ensemble). *Text only*: finished rule + `chk` for every model. A model left with no feature falls back to `chk`. Modified recipes use threshold 0 (S1 at threshold 0: 32 / 1.00). Output features carry most of the signal; of the internal readouts, only S2's Olmo direction changes the totals.

Variant	Qwen	Skywork	Olmo	DeepSeek	MAIN	LB
All robust	8	9	7	7	31	0.50
Text only	10	9	8	7	34	0.57
Text + output (no internals; S1 = S2)	9	10	6	8	33	1.00
S1: text + internal (no output)	8	9	6	7	30	0.71
S2: text + internal (no output)	8	9	8	7	32	0.79
S1 (all three)	9	10	5	9	33	1.00
S2 (all three)	9	10	8	9	36	0.93

is 72% robust, S2 beats all-robust by 5 calls and S1 by 2, within the intervals, and the text-only variant (finished rule and check share) already reaches 34/43 (Table 3). The two submissions trade places across the two label sets, which is why we nominate both. Among all builds we scored on the leaderboard (Appendix E), the per-model builds scored 0.64–1.00 and single-family builds 0.28–0.71.

5 Ablations

Components (Table 2). The regime split is not a predictor on its own but settles 11/43 MAIN rows at zero error. The Olmo router trades two MAIN rows for two leaderboard problems; the judge makes the same leaderboard calls as the router. S1 keeps the router and judge for Olmo and is the hedge against the Olmo direction of S2, which rests on only 4 unfinished MAIN rows.

5.1 Does the white-box signal matter?

The internal signals are DeepSeek's prompt-attention feature in both submissions, plus Olmo's attention-based familiar branch and Qwen judge in S1 and Olmo's hidden-state direction in S2; Qwen's own calls use H_{4096}, an output feature, in both. In S1 the internals make no measurable difference: without them the recipe also scores 33/43 and 1.00, because Olmo's unfinished calls are decided mostly by the Min-K% router, an output feature, and the attention-based familiar branch calls almost every problem it receives robust. The internal readout that matters is S2's hidden-state direction, which adds 3 MAIN rows for one leaderboard problem and gives the best total over both sets (49/57, against 47 without internals). Output features carry most of the signal: without them leaderboard accuracy falls to 0.71–0.79. Recalibrating every variant's thresholds label-free on the unlabelled pool moves individual cells by up to 3 calls but leaves these conclusions unchanged. Appendix B removes one feature at a time.

Table 4: Where in the trace to read the signal: AUC on unfinished traces (robust = positive class, oriented by the shipped sign) on our labels / MAIN. Left: Qwen forced-answer entropy at truncation b (MAIN: 7 R / 4 NR). Right: Olmo direction projections by trace position (MAIN: 2 R / 2 NR). Olmo's "ours" AUCs are in-sample, since the direction was fitted on those labels. On our labels the entropy signal grows with b; on MAIN every budget is similar, so the exact budget matters little.

Qwen H_b, $b =$	ours	MAIN
512	0.44	0.75
1024	0.49	0.64
2048	0.61	0.71
3072	0.69	0.75
4096	0.77	0.71
6144	0.79	–

Olmo direction at	ours	MAIN
25% of trace	0.99	1.00
50% of trace	0.97	1.00
75% of trace	0.93	1.00

Truncation point and position (Table 4). On our labels the forced-answer entropy becomes informative only after about 2048 tokens of reasoning; on MAIN all budgets give 0.64–0.75, so the choice of 4096 (made with MAIN and earlier leaderboard scores in view) matters little. For Olmo, the direction separates MAIN's four unfinished rows at every position, so 25% is not supported over 50% or 75%; this is the weakest-supported component, and the reason S1 is nominated alongside S2.

Alternatives tried (Appendices D and E). We also tested whole-trace hidden-state geometry (K-means walk graphs, δ-hyperbolicity and Poincaré embeddings [Raj, 2026], CKA [Kornblith et al., 2019]), attention-map spectra [Bińkowski et al., 2025], Min-K%++ [Zhang et al., 2025], kNN and TDA trajectory features, PEARL-style prompt noising, and sentence-embedding reasoning graphs. Several reach high AUC on one label set but none holds up across our labels and MAIN; builds based on single families scored 0.28–0.71 on the leaderboard.

6 What do the official labels measure?

We tried to enlarge the labelled data by applying the official rule ourselves, and found two problems with the resulting labels.

Noise and difficulty. With 8 samples per version, a problem solved about half the time can cross the 0.25 drop threshold by chance. Our first labelled problems were also much harder for the models than MAIN's: on unfinished traces, the median forced-answer top-1 probability at 6144 tokens was 0.54 against 1.00 for Qwen, and 0.92 against 1.00 for Skywork. We then labelled a second set of 109 problems matched to MAIN's confidence (forced top-1 probability at least 0.7). It closed the difficulty gap but not the noise: we obtained 10, 10 and 8 non-robust labels for Qwen, Olmo and Skywork, about as many as a no-effect null predicts (9.3, 10.3 and 7.1). Relabelling 62 Skywork problems at 32k with natural answers left 8 non-robust cases; our Skywork recipe was at chance on them (balanced accuracy 0.52), and original-problem accuracy still predicted the label (AUC 0.74).

Non-reproduction. Our perturbations also miss official non-robust cases. MAIN problem `7ee9d443` is officially non-robust for all four models (drops 0.40–1.00), yet under our perturbations Qwen and Skywork solve it in 47–48 of 48 samples at a 32k cap, in 67–68 of 68 across 30 further rewordings, and also with changed numbers and other prompt formats. Our sampling settings match the official ones, so the official labels must depend on perturbation texts or grading details we do not have.

Consequence. Such labels can still fix the sign of a consistent feature, but cannot measure accuracy. We therefore evaluate only on MAIN and the leaderboard, and keep each choice made on our labels (Section 2) only if it also holds on the official ones.

Limitations

Small, partly in-sample evaluation. MAIN's 43 rows give intervals about ± 0.12 wide; S1 and S2 differ by 3 MAIN calls and one leaderboard problem, neither of which is significant. MAIN is not fully held out: feature signs for the shared features used MAIN (length-controlled AUC), reference sets include MAIN's unfinished values, and both single-feature readouts, as well as the per-model subsets of later builds, were chosen using MAIN results and the leaderboard scores of earlier builds.

Unreliable self-generated labels (Section 6) leave the signs and the Olmo direction resting on 21–66 noisy labels per model. **Distribution shift.** Features calibrated on our data transferred unevenly to leaderboard scores (single-family builds 0.28–0.71), so accuracy on the private 43 problems may differ substantially from MAIN. **Replay gap.** Olmo's judge was replayed on only 2 of the 4 unfinished MAIN rows. **Greedy-only view.** We observe one greedy trace capped at 6144 tokens, while the labels come from temperature-1 samples with a much larger budget.

AI usage statement

We used AI assistants extensively. **Claude Code, mainly with Claude Opus 5.5 (Anthropic),** was used for much of the code and experimentation: it wrote most of the analysis, feature-extraction, labelling and submission code under our direction; ran and monitored the GPU experiments. **Claude Sonnet** (through claude.ai) generated the perturbation texts for our own labelled data, following a written specification (Section 2). **Gemini Deep Research** (Google) and **Claude Sonnet** assisted the literature review, suggesting candidate papers and methods; we read and checked every paper we used. The authors chose the research directions, decided every submission and nomination, checked the reported numbers against the experiment logs, and are responsible for all claims. No AI system is an author.

References

Alexandra Bazarova et al. Hallucination detection in LLMs with topological divergence on attention graphs, 2025.

Jakub Bińkowski, Denis Janiak, Albert Sawczyn, Bogdan Gabrys, and Tomasz Kajdanowicz. Hallucination detection in LLMs using spectral features of attention maps, 2025.

Paul C. Bogdan, Uzay Macar, Neel Nanda, and Arthur Conmy. Thought anchors: Which LLM reasoning steps matter?, 2025.

Yung-Sung Chuang, Linlu Qiu, Cheng-Yu Hsieh, Ranjay Krishna, Yoon Kim, and James Glass. Lookback lens: Detecting and mitigating contextual hallucinations in large language models using only attention maps. In *Proceedings of EMNLP*, 2024. arXiv:2407.07071.

DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning, 2025.

Djiré, Abdoul Kader Kaboré, Earl T. Barr, Jacques Klein, and Tegawendé F. Bissyandé. Memorization or interpolation? detecting LLM memorization through input perturbation analysis, 2025. Introduces PEARL.

Zenghao Duan, Liang Pang, Zihao Wei, Wenbin Duan, Yuxin Tian, Shicheng Xu, Jingcheng Deng, Zhiyi Yin, and Xueqi Cheng. Circular reasoning: Understanding self-reinforcing loops in large reasoning models, 2026.

Minju Gwak, Minseo Kwak, Dongseok Lee, Guijin Son, Alan Ritter, and Jaehyung Kim. LaRA: Layer-wise representation analysis for detecting data contamination in RL post-training, 2026.

Jujie He et al. Skywork open reasoner 1 technical report, 2025a.

Zirui He, Haiyan Zhao, Ali Payani, and Mengnan Du. MemLens: Uncovering memorization in LLMs with activation trajectories, 2025b.

Hazel Kim, Tom A. Lamb, Adel Bibi, Philip Torr, and Yarin Gal. Detecting LLM hallucination through layer-wise information deficiency: Analysis of ambiguous prompts and unanswerable questions, 2024.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning (ICML)*, 2019.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023.

Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, et al. Measuring faithfulness in chain-of-thought reasoning, 2023.

Yang Li, Zhichen Dong, et al. Attention illuminates LLM reasoning: The preplan-and-anchor rhythm enables fine-grained policy optimization, 2025.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

Iman Mirzadeh, Keivan Alizadeh, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. GSM-Symbolic: Understanding the limitations of mathematical reasoning in large language models. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.05229.

nostalgebraist. Interpreting GPT: the logit lens. LessWrong blog post, 2020.

Qwen Team. Qwen3.5-4B, 2026. Model card, `https://huggingface.co/Qwen/Qwen3.5-4B`.

Arnav Raj. Hyperbolic geometry of reasoning: Probing LLM hidden states. In *ICLR 2026 Workshop on Geometry-grounded Representation Learning and Generative Modeling (GRaM), Tiny Paper Track*, 2026.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP*, 2019.

Xu Shen, Zhen Tan, Song Wang, Pingjun Hong, Rui Miao, Xin Wang, and Tianlong Chen. Detecting unfaithful chain-of-thought via circuit-guided internal-external discrepancy, 2026.

Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, Daogao Liu, Terra Blevins, Danqi Chen, and Luke Zettlemoyer. Detecting pretraining data from large language models. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.16789.

Michal Stefánik, Philipp Mondorf, Andreas Waldis, Qianying Liu, Chuan Yang, Michal Spiegel, Josef Kuchar, Marek Kadlcík, Adam Vawda-Oomerjee, Chaoran Liu, Simon Frieder, Barbara Plank, Fazl Barez, and Pontus Stenetorp. AIMO interpretability challenge. *CoRR*, abs/2607.13899, 2026. doi: 10.48550/ARXIV.2607.13899. URL `https://doi.org/10.48550/arXiv.2607.13899`.

Xue Wen Tan, Nathaniel Tan, Galen Lee, and Stanley Kok. The shape of reasoning: Topological analysis of reasoning traces in large language models, 2025.

Team Olmo. Olmo 3, 2025. Model card, `https://huggingface.co/allenai/Olmo-3-7B-Think`.

Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, et al. Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP: System Demonstrations*, 2020.

An Yang et al. Qwen3 technical report, 2025.

Fan Yin, Jayanth Srinivasa, and Kai-Wei Chang. Characterizing truthfulness in large language model generations with local intrinsic dimension. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.18048.

Jingyang Zhang, Jingwei Sun, Eric Yeats, Yang Ouyang, Martin Kuo, Jianyi Zhang, Hao Frank Yang, and Hai Li. Min-K%++: Improved baseline for detecting pre-training data from large language models. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2404.02936.

A Feature definitions

All features are computed on the prompt p and the greedy trace t (at most 6144 tokens), with suffix σ = "`\n</think>\n\nThe final answer is $\boxed{`".

- `chk`: share of the thinking's sentence units ($\geq$4 words) that match a check/verify/confirm lexicon; undefined below 10 units.
- `fa`: $\max_v P(v \mid p, t_{1:6144}, \sigma)$.
- `nrk`, `f1k`: with a^* the clean forced answer token, $P(a^* \mid p, \sigma)/P(a^* \mid p, t, \sigma)$ and $P(a^* \mid p, t_{1:1024}, \sigma)/P(a^* \mid p, t, \sigma)$.
- `cfc`, `cfl`: for each automatic rewording p' (typos at rate 0.1, a fixed distractor sentence after the first sentence, single-letter variables in math renamed consistently), teacher-force t after p'. `cfc` is the mean cosine between original and reworded 16-token-window hidden states over the last quarter (averaged over 6 layers); `cfl` is the mean change in per-token log-likelihood of t over its first tokens.
- `lble`, `pmle`: per 16-token span, all layers and heads, the Lookback-Lens ratio (attention on the prompt over attention on prompt plus generated tokens) and the raw prompt attention mass; mean over the last quarter of spans minus the first quarter.
- `jmp`: on the 64-token block graph (each block's attention on earlier blocks, normalised per block), the mean attention-weighted backward distance in blocks.
- H_b: entropy of $P(\cdot \mid p, t_{1:b}, \sigma)$.
- seg-25% direction: $\langle \bar{h}^{(5)}_{[0.25L \pm 128]}, d \rangle$, with $\bar{h}^{(5)}$ the mean 5th-layer state over the 256 tokens around position $0.25L$, and d the normalised difference of class means on our labelled unfinished Olmo traces.
- Min-K% ($k = 20$): mean of the lowest 20% token log-probabilities of the problem text under the model.

B Drop-one-feature ablation

Table 5 removes one feature at a time from S1's recipe, keeping the shipped thresholds; for Olmo, features are removed from the router branch that uses them. Scores are correct calls on MAIN (43 rows) and the leaderboard (14).

Table 5: Drop-one-feature ablation of S1 (33 / 14). Group: T = text, O = output distribution, I = internal. Every Skywork and DeepSeek feature carries weight, including DeepSeek's prompt-attention feature (2 MAIN rows); removing any feature from Olmo's familiar branch gains one MAIN row at no leaderboard cost.

Model	Feature removed	Group	MAIN (43)	LB (14)
—	none (S1)	—	33	14
Skywork	`fa`	O	32	14
Skywork	`chk`	T	33	13
Skywork	`f1k`	O	33	12
DeepSeek	`chk`	T	32	13
DeepSeek	`nrk`	O	31	14
DeepSeek	`cfl`	O	31	14
DeepSeek	`pmle`	I	31	14
Olmo, familiar branch	`chk`	T	34	14
Olmo, familiar branch	`lble`	I	34	14
Olmo, familiar branch	`pmle`	I	34	14
Olmo, novel branch	`f1k` (replaced by `chk`)	O	35	12
Qwen	H_{4096} (back to the rank ensemble)	O	32	13

C Per-feature evidence

Table 6 gives each shipped feature's own AUC on unfinished traces, oriented by the sign the recipe uses, so values above 0.5 mean the feature points the way it is used. The columns are our first labelled problems (the signs were chosen there, so these are in-sample), a second labelled set that we drew to match MAIN's forced-answer confidence and used for no choice, and MAIN. All use forced labels at 6144 tokens, and no length control is applied. The MAIN counts are small (Olmo 2 + 2), so a single row moves an AUC by up to 0.25. On the first labelled set, every feature except Skywork's

Table 6: AUC of each shipped feature on unfinished traces, oriented by its shipped sign. n = robust / non-robust traces. DeepSeek has no confidence-matched set. Olmo's `f1k` is shown with the novel branch's sign; the familiar branch does not use it.

Model	Feature	Used as	Sign	Ours, first	Ours, matched	MAIN
Qwen	H_{4096}	S1/S2 readout	−	0.77	0.50	0.71
	`lble`	fallback ensemble	+	0.82	0.58	0.61
	`pmle`	fallback ensemble	+	0.80	0.65	0.50
	`jmp`	fallback ensemble	+	0.80	0.46	0.89
	`cfc`	fallback ensemble	−	0.69	0.48	0.43
	n			43 / 23	25 / 10	7 / 4
Skywork	`fa`	ensemble	+	0.68	0.29	0.75
	`chk`	ensemble	+	0.76	0.40	0.50
	`f1k`	ensemble	−	0.43	0.09	0.75
	n			19 / 12	16 / 8	4 / 3
Olmo	`f1k`	novel branch	+	0.39	0.64	1.00
	`chk`	familiar branch	+	0.72	0.45	0.50
	`lble`	familiar branch	+	0.56	0.58	0.50
	`pmle`	familiar branch	+	0.68	0.71	0.50
	seg-25% direction	S2 readout	+	0.99	0.39	1.00
	n			12 / 9	17 / 10	2 / 2
DeepSeek	`chk`	ensemble	+	0.71	—	0.67
	`nrk`	ensemble	−	0.63	—	1.00
	`cfl`	ensemble	−	0.65	—	0.62
	`pmle`	ensemble	+	0.68	—	0.05
	n			27 / 10	—	7 / 3

`f1k` and Olmo's novel-branch `f1k` points the shipped way. Both of those signs follow MAIN rather than our labels. On the held-out confidence-matched set, 8 of the 13 features are at or below 0.5, and Skywork's three features reverse (0.09–0.40), the pattern Section 6 describes. On MAIN, 10 of 17 are above 0.5, 5 are exactly 0.5 and 2 are below. DeepSeek's `pmle` reverses there (0.05), yet removing it costs 2 MAIN rows (Table 5): its value in the ensemble comes from how it combines with `nrk` and `cfl`, not from its own ranking. The Olmo direction's 0.99 on our first set is in-sample, because the direction was fitted there.

D Exploration: what else we tried

The submissions are the end of about 120 numbered experiments and 48 leaderboard builds (Appendix E). This appendix summarises the feature families we explored and why most were dropped, with emphasis on the white-box ones. Unless stated otherwise, "AUC" is the robust-vs-non-robust AUC on MAIN after controlling for trace length (robust = positive, so values below 0.5 mean the feature points the other way), and "our labels" means nested cross-validation (feature or layer selected inside each training fold) on the labelled sets of Section 2, reported as accuracy on forced / natural labels. Section 6 explains why results on our labels are weak evidence.

How the exploration went. We first screened features on a large pool of organiser-provided permutations, then found its non-robust labels were dominated by permutations that remove information from the problem, and moved to MAIN with length controls. Main's 43 rows could rank feature families but not validate ensembles: with the feature choice nested inside leave-one-problem-out, fixed mixes that looked near-perfect in-sample fell to 0.56–0.77. We then labelled our own problems to get more

rows, ran three GPU feature batteries over them (262–518 scalar features per trace, plus per-layer and per-head arrays), and from the per-model builds onward chose recipes with MAIN and leaderboard scores in view. The equal-weight percentile mean of Section 3 was never beaten held out by cascades, stacking or random forests; the last two fell below chance on MAIN.

Hidden states.

- *Prompt-token probe* (linear probe on the last prompt token, a few layers): below chance per model on MAIN (0.33). A ridge probe on the same state predicts thinking length (cross-validated R^2 0.57 on the organiser pool) but carries no robustness signal of its own.
- *Difference-of-means directions* per model and layer, fitted leaving one problem out: the right sign at every layer (0.57–0.67) but never beyond a permutation null. Mean states of fixed trace segments did better, 0.82 on MAIN for the segment at 25% of thinking; a difference-of-means direction on that segment, fitted on our labels, became S2's Olmo direction (Section 3.5).
- *Trajectory geometry* of the residual stream over time (tortuosity, speed, late-vs-early cosine, on 16-token waypoints and 16/64/256-token moving averages): mostly explained by length; late-vs-early cosine kept 0.78 after controls.
- *Anchoring*: cosine between the current reasoning state and the problem's state. Inverted on MAIN (0.30).
- *Step similarity*, prompted by work on self-reinforcing reasoning loops [Duan et al., 2026]: cosine between consecutive 64-token states. More similar steps go with non-robust (0.21–0.32 after averaging over chunk phases).
- *Reasoning as a state graph*, following the cluster-graph construction of Duan et al. [2026]: K-means states (K = 16, 32, 64) turn each trace into a walk; cosine kNN graphs on averaged trajectories. Robust traces form globally integrated graphs (kNN global efficiency 0.78; K-means cycle rank and spectral gap up to 0.86). The K-means features are at chance by the number of hits and vanish once step similarity is removed; the kNN signal survives that but holds in only three of the four models, and it is not stable across traces of the same problem: on MAIN, with one greedy and two temperature-1 traces per problem, its AUC is 0.82 / 0.50 / 0.45 and its rank correlation across the three traces is 0.06. Every leaderboard build that included it scored 0.28–0.50, except one in which it only broke ties.
- *Topological data analysis* [Tan et al., 2025] (Vietoris–Rips persistence on trajectory nodes; per layer and in sliding windows): at chance overall (7 of 214 tests at $p < 0.05$).
- *δ-hyperbolicity, compression and Poincaré embeddings* [Raj, 2026]: the best DeepSeek feature on our labels (δ at 50% depth, 0.81 / 0.79), but the hyperbolic map never beat plain geometry.
- *CKA* [Kornblith et al., 2019] across depth, and between clean and perturbed teacher-forced passes: weak; best Olmo cosine drift under code typos 0.79 / 0.76 on our labels.
- *Recall-vs-computation directions*, trained on separate recall and computation prompts: significant at the null rate on our labels.
- *Late-layer MLP write size* at the forced-answer position: larger writes go with non-robust on both MAIN (0.25) and our olympiad labels (0.31), consistent with a recalled answer injected late; it was not shipped.
- *Logit lens* [nostalgebraist, 2020] *on the answer digits*, following MemLens [He et al., 2025b] (where the answer digit emerges across layers, a memorisation signal): fully explained by whether the trace finished.

Attention.

- *Laplacian spectra of attention maps* [Bińkowski et al., 2025]: at or below chance as summary features on MAIN; as a trained probe, 0.73 on our olympiad labels and 0.83 when transferred to MAIN (22 rows). Selecting a single head inside each fold gave Olmo 0.92 / 0.85 on our labels; none of these was shipped.
- *Lookback Lens* [Chuang et al., 2024]: the paper's per-head logistic regression fails at our sample size (0.40–0.44 held out). Its late-minus-early summary became `lble`, and the same summary of prompt attention became `pmle`.

- *TOHA* [Bazarova et al., 2025] (minimum spanning forest from response tokens to the prompt, a faithful port with head selection nested): inverted on MAIN for the answer window (0.20); held out per model at most 0.66 (DeepSeek).
- *Attention periodicity and layer-to-layer change* (our own features: FFT over positions; KL between adjacent layers): nothing on MAIN.
- *Block jump*: attention-weighted backward distance between 64-token blocks, 0.70 on unfinished rows and not explained by looping or repetition (each $|\rho| \leq 0.14$). Shipped as `jmp`.

Interventions and perturbations.

- *Activation patching* [Meng et al., 2022] *and knockouts* (patching the no-reasoning run into the full trace; knocking out the problem tokens): the best features (0.67) duplicate forced-answer confidence and add nothing to it.
- *Rewording sensitivity* under three automatic rewordings, teacher-forcing the original trace: hidden-state cosine and log-likelihood change were shipped as `cfc` and `cfl`.
- *PEARL-style prose noise* [Djiré et al., 2025] (memorised inputs should break abruptly under rising input noise; we used letter typos at 2–30% in the prose only, never in numbers or math, and curves over the noise rate): mean-state drift area 0.77 on MAIN, but nothing in nested combinations.
- *A direct test*: regenerate the answer for five perturbations and call robust if all agree. Agreement AUC 0.81 for Skywork on our olympiad labels, but five extra long generations do not fit the per-call time limit.

Text and output distribution. Reasoning markers (hesitations, restarts) and the loop statistics of Duan et al. [2026] were null after length controls, except check share, which was shipped as `chk`. Sentence-embedding [Reimers and Gurevych, 2019] topic graphs reached 0.78 on MAIN (spectral gap; robust traces stay on fewer topics) but did not replicate on our 32k relabel. Forced-answer confidence and its slope over budgets were the strongest single features on MAIN (0.90 at 6144 tokens). Min-K% and Min-K%++ [Shi et al., 2024, Zhang et al., 2025] familiarity scores became the Olmo router.

Ports of recent papers. We also implemented seven recent methods as faithfully as our setting allows, checking each against its paper, and scored them on our earlier labelled unfinished rows (88 rows; 7% of tests at $p < 0.05$ against 5% by chance):

- *Layer-wise usable information* [Kim et al., 2024]: the strongest, 0.67 with the same direction in all four models, but correlated with forced-answer confidence and check share (ρ 0.41–0.57; 0.56 after removing them).
- *Local intrinsic dimension* [Yin et al., 2024]: 0.54 pooled, 0.77–0.80 for Skywork only.
- *LaRA* representation collapse [Gwak et al., 2026]: model-specific, from 0.08 to 0.65.
- *Attention rhythm* [Li et al., 2025], *receiver heads* from Thought Anchors [Bogdan et al., 2025], Min-K%++ [Zhang et al., 2025] and a circuit-guided internal–external discrepancy [Shen et al., 2026]: at chance. Thought Anchors' attention suppression pointed the wrong way (0.30) and not consistently across models.

E Builds scored on the leaderboard

Table 7: Leaderboard accuracy of our builds with their MAIN correct calls (each build's own recipe replayed on our MAIN traces). Per-model builds, the family S1 and S2 belong to, score highest on both.

Build	Leaderboard	MAIN (43)
v8-ent4096 (S1)	1.00	33
v8-ent4096-olmoseg25 (S2)	0.93	36
v8 (base recipe, no single-feature readouts)	0.93	32
v7, v5 (per-model, router added in v7)	0.93, 0.86	32, 34
v6 (per-model, before the Olmo router)	0.93	34
max (per-model)	0.86	33
lookback (per-model)	0.71	34
cascade, v4, ensemble (per-model)	0.64	30–34
single-family builds (24)	0.28–0.71	19–34

F Reproducibility

Code, recipes (`recipe.json`), shipped artifacts and the analysis scripts are at `https://github.com/RKResident/AIMO-Interp-Challenge`: `submission_core/` (inference), `analysis/build_submission_artifacts.py` (calibration: references, thresholds, per-model recipes), `analysis/build_datasets.py` (our labelled datasets, `datasets/`) and `reproduce/`. Running `python reproduce/replay.py --main <train-main-v2 parquet>` regenerates every MAIN number in Tables 1–4 and Appendices B–C on CPU in seconds. It uses the recipes inside the submitted zips, the shipped scoring code and our per-trace feature values (`reproduce/features.csv`); the README gives the details. The remaining analysis scripts are published for transparency and depend on intermediate data that is not released. Feature extraction and labelling ran on one NVIDIA RTX PRO 6000 and one A10G, using vLLM [Kwon et al., 2023] for labelling samples and HF Transformers for features. Model licences: Qwen3.5-4B, Skywork-OR1-Math-7B and Olmo-3-7B-Think are released under Apache-2.0, DeepSeek-R1-0528-Qwen3-8B under MIT. The challenge data are used under the competition's terms and are not redistributed.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 state results only on the official labels, with in-sample caveats (Section 4 and the Limitations section).

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: See the Limitations section after the main text.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

Justification: The paper has no theoretical results.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Section 3 and Appendix A define every feature, threshold and decision rule; the shipped `recipe.json` files hold all values.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
     (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Code, recipes and analysis scripts are released (Appendix F); the official challenge data are not ours to redistribute.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Sections 2–3: data splits, how signs, thresholds and recipes were chosen; nothing is trained by gradient descent.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Table 1 gives Wilson 95% intervals; Section 4 and the Limitations note that the two submissions do not differ significantly.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).

- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Per-problem cost in Section 3; hardware in Appendix F.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The work uses public models and challenge data and involves no human subjects.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [N/A]

    Justification: The work is a diagnostic study of public models' reasoning robustness with no direct deployment path or foreseeable negative societal impact.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.

- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: We release no data or models with a high risk of misuse.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: Models, libraries and methods are cited; licences are listed in Appendix F.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.

- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The released code and recipe artifacts are documented in the repository (Appendix F).

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: No crowdsourcing or human-subject research.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: No human subjects.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: An LLM (Claude Sonnet) wrote the perturbation texts for our own calibration labels, and a second open model (Qwen) acts as a judge for Olmo (Section 3); all LLM use is described in the AI usage statement.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.